

SIEM Dashboards & Alerts

SOC Analyst Internship — Week 4, Task 12

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Task	Week 4 - Task 12: SIEM Dashboards & Alerts
Tool Used	Splunk Enterprise (Free Tier)
Deadline	September 14, 2026, 10:00 PM (Pakistan Time)

Objective

Make the SIEM useful by building a dashboard to visualize authentication activity and creating an alert that triggers automatically when suspicious activity (multiple failed logins) is detected.

1. TryHackMe Room Completed

The "Introduction to SIEM" room on TryHackMe was completed in full (100%), covering SIEM fundamentals, log sources, alerting process, and hands-on lab work across all 7 tasks.

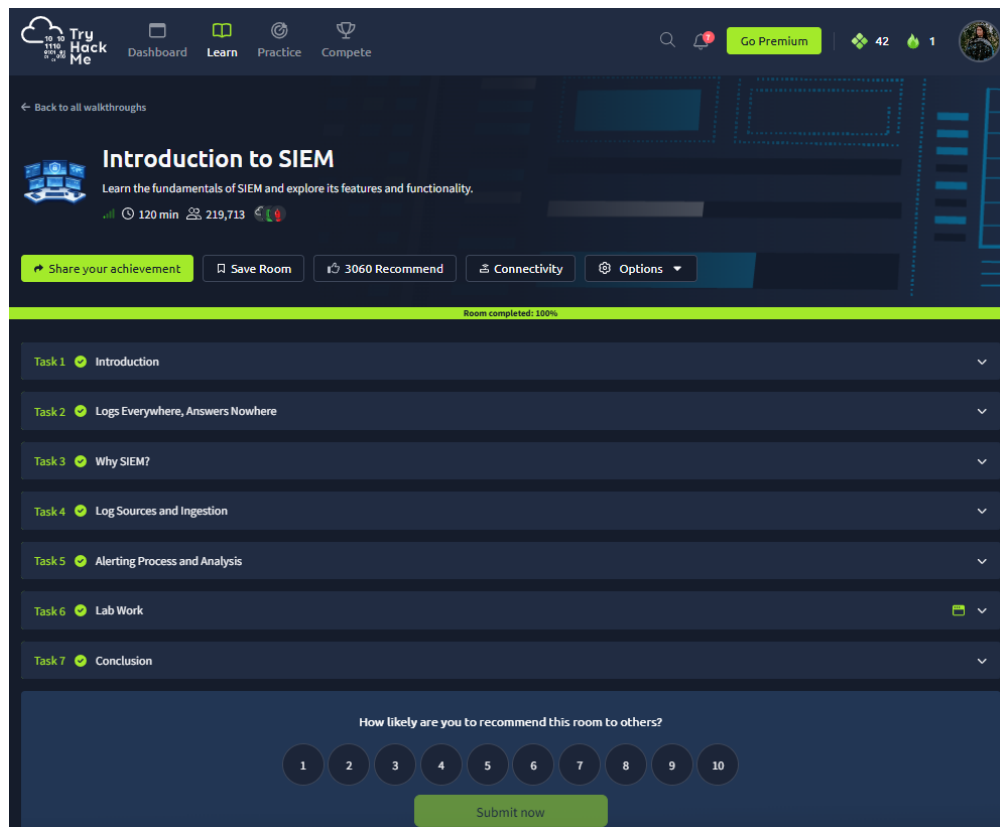


Figure 1: TryHackMe – Introduction to SIEM room, 100% completed

2. Search to Visualization & Dashboard

A sample authentication log (sample_auth.log) was ingested into Splunk. The log contains login/logoff events with fields such as user, action, status, and src_ip.

Search queries used:

- Failed logins over time: sourcetype="auth_log_fixed" status=failed | timechart count
- Successful logins by user: source="sample_auth.log" "success" | stats count by user

Both searches were converted into visualizations (column charts) and saved as panels on a single dashboard named **"SOC Monitoring Dashboard."** The dashboard contains two panels as required: *Failed Logins Over Time* and *Successful Logins by User*.

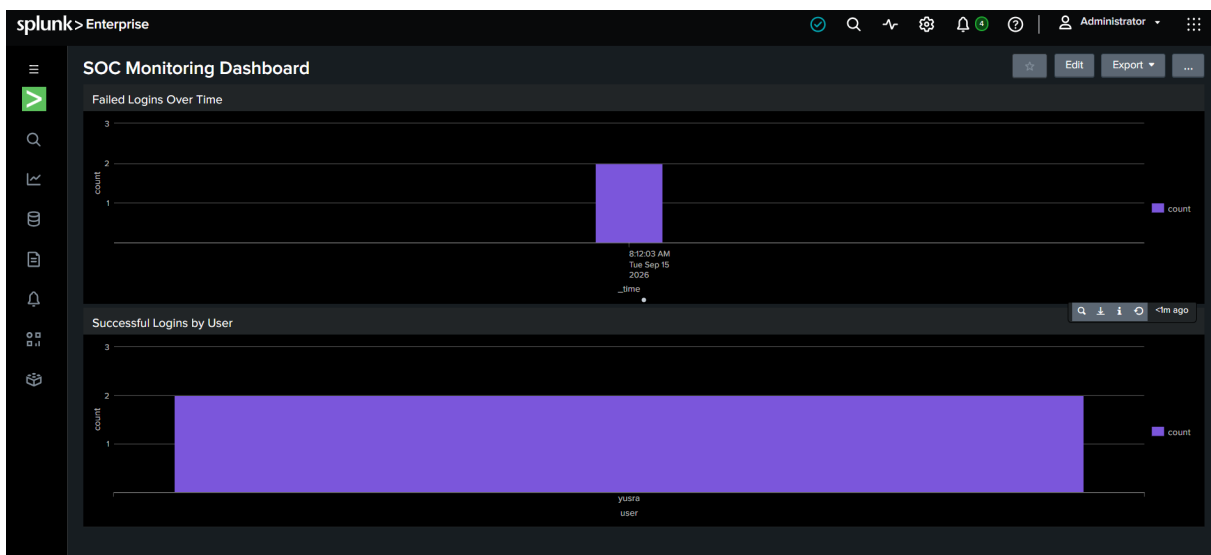


Figure 2: SOC Monitoring Dashboard with 2 panels — failed logins over time and successful logins by user

3. Alert Creation

An alert named **"High Failed Login Alert"** was created to detect suspicious login activity automatically.

Alert configuration:

Search Query	sourcetype="auth_log_fixed" status=failed
Alert Type	Scheduled - Cron Schedule (* / 5 * * * *, every 5 minutes)
Trigger Condition	Number of Results is greater than 5
Trigger Action	Add to Triggered Alerts
Time Range	All time
Expires	24 hours

The underlying log data contained 12 failed login events across three different users (admin, administrator, guest), which exceeds the configured threshold of 5, ensuring the alert condition would be met on the next scheduled run.

4. Alert Testing & Trigger Proof

The alert was left running on its 5-minute cron schedule and verified against the Trigger History log on the alert's detail page. The alert fired successfully and repeatedly, confirming that the SIEM correctly detects and flags suspicious activity in real time.

The screenshot shows the Splunk Enterprise interface. On the left is a navigation sidebar with options like Search & Reporting, Search, Analytics, Datasets, Reports, Alerts, Dashboards, and Modules. The main content area is titled 'High Failed Login Alert' and includes the following details:

- Search Query:** `sourcetype="auth_log_fixed" status=failed`
- Alert Type:** Scheduled - Cron Schedule (* / 5 * * * *, every 5 minutes)
- Trigger Condition:** Number of Results is greater than 5
- Trigger Action:** Add to Triggered Alerts
- Time Range:** All time
- Expires:** 24 hours

Below these details is a 'Trigger History' section with a table showing 3 successful fires:

	TriggerTime	Actions
1	2026-09-16 22:05:00 Pakistan Standard Time	View Results
2	2026-09-16 22:00:01 Pakistan Standard Time	View Results
3	2026-09-16 21:55:01 Pakistan Standard Time	View Results

Figure 3: Alert detail page showing Enabled status, trigger condition, and Trigger History with 3 successful fires (22:05, 22:00, 21:55 PKT)

Trigger history log:

#	Trigger Time (Pakistan Standard Time)
1	2026-09-16 22:05:00
2	2026-09-16 22:00:01
3	2026-09-16 21:55:01

5. Summary

- Completed the TryHackMe "Introduction to SIEM" room (100%).
- Converted a Splunk search into a visualization (column chart).
- Built a dashboard with 2 panels: Failed Logins Over Time, Successful Logins by User.
- Created an alert ("High Failed Login Alert") that triggers when failed logins exceed 5, scheduled every 5 minutes.
- Tested the alert against real log data (12 failed logins) and confirmed it fired 3 times consecutively, verified via Trigger History.

Challenges & Notes

Initially, the uploaded log file was indexed as a single merged event instead of 20 separate events, which caused search counts to be inaccurate. This was resolved by re-uploading the file with `SHOULD_LINEMERGE=false` and a custom `LINE_BREAKER` setting under a new sourcetype (`auth_log_fixed`), after which each log line was indexed as its own event and search/alert counts became accurate.